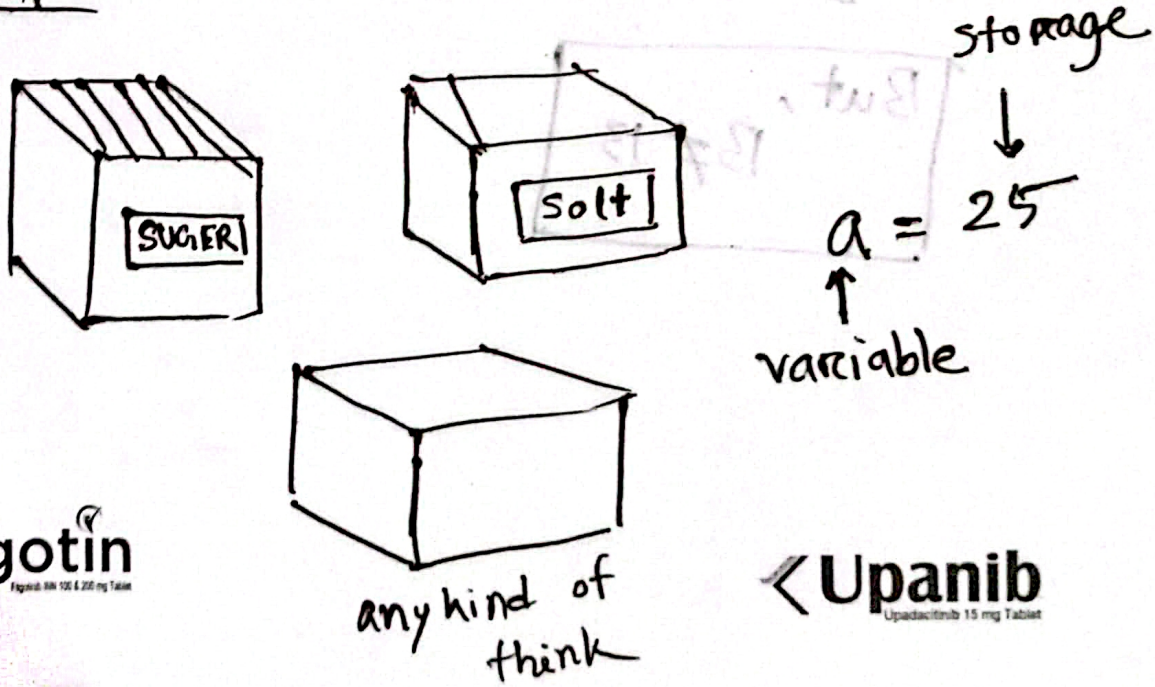
```
#include <stdio.h>
int main() {
  printf("Hello world");
  return 0;
}
```

4. The Foundation: Variables and Memory

> a varciables is like a food containen; a named location to store your data

Example



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Case Sensitive

A = □
a = □

A = 35
a = 35 ✗

A ≠ a

Problem

A = 13
B = 6

now, variables

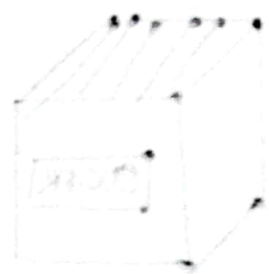
A = 13
B = 6

A = B

A = 6

B = A = 6

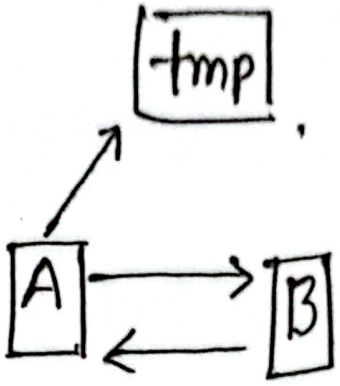
But, B ≠ 13

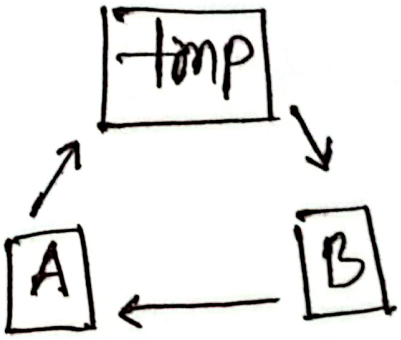


A=13 B=6

tmp = A,

tmp=13





A=B

A=6

B = tmp

B = 13

now,

A = 6

B = 13

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The Fixed Valus (constants)

Integer: 1, 0, -2. (Int),

Real: 1.0 , 3.14 , -24.0

Character: 'a', 'A', '#' (Char)

Valid Naming

age
_total
my_age
StudentName
TotalScore
value1
__init

Used: A-Z } start program
a-z
1-9
abcarc

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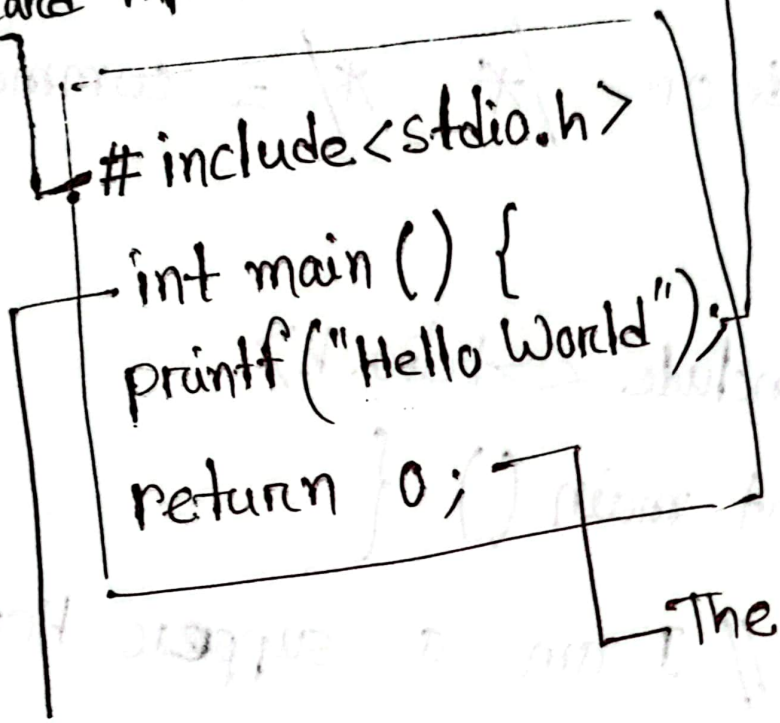
Compiler Errors

1st_num	First character must be an alphabet letter
my age	Don't spaces
total$	Don't use symbol
int	
email@	
2nd_place	

The Instruction: The action you want the computer to perform

The Header: Imports the standard input/output library



The exit status

The Entry Point: Where the compiler begins reading your logic

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The Director's Notes: Using comments

// /* */

// or /* , */ = comments

```
# include <stdio.h>
int main () {
// I am a supper Hero
return 0; }
```

comments don't show program

limit 1 line

whenever // /* */ unlimit

Data types → int, float, charc

~~Spatial footprint Matrix~~

int → integerc -

float → $\frac{7}{2} = 3.5$

Charc →

int = storage for 10 deget

10 deget ↑↑ = we use to long

char / signed char / unsigned char.

↳ 2 to 4 Bytes

int / short int (2 bytes)

float / long int (4 bytes)

(i) double (8 bytes)

(ii) long double (10 bytes)

concept	Date type	Translation key
Integers (1, 0, -2)	int	%d
Real Numbers (3.14, -24.0)	float/double	%f
characters ("a", "#")	char	%c

```
#include <stdio.h>
int main () {
    int a = 5;
    int b = 7;
    printf ("%d", a);
    return 0;
}
```

```
int a = 5;
int b = 7;
printf ("%d %d", a, b);
```

scanf ("%d, &age);

Operators

Arithmetic
+ - * / %

Bitwise
& | ^

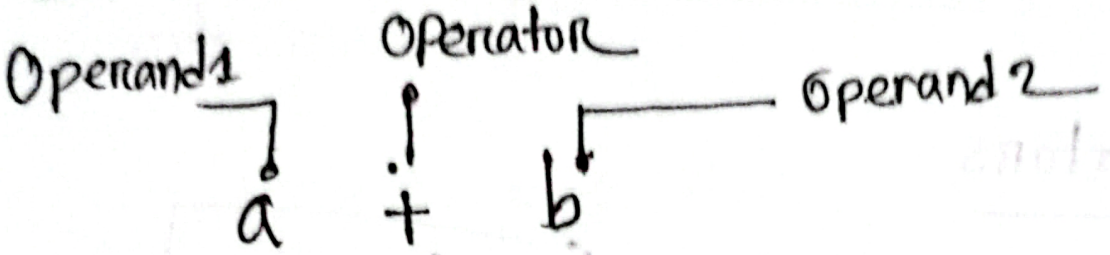
Relational
> < ==

Assignment
= += -=

Logical
&& || !

Ternary

Arithmetic Operands




```
# include <Stdio.h>
  int main () {
  int a = 25;
  int b = 26;
  int product = 25 * 26
      printf("%d", product);
  return 0;
  }
```

only for veriable

int product = a * b

output : 650

```
int a = 30;
int b = 30;
int sum = a+b;
printf ("%d", sum);
```

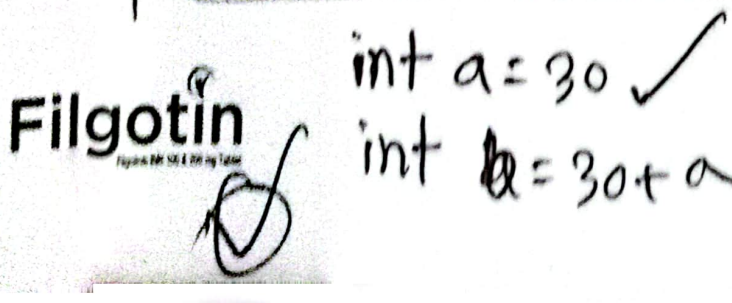
output = 60

Short cut:

```
int a =30;
int b=30;
printf ("%.d", a+b);
```

output = 60

The compiler evaluates everything on the right and pours the final result into the single container.



valid

$a = b + c$

$a = b * c$

$a = b / c$

Modulo (%)

Division (/)

$$2 \overline{)\,3}\ 1 \quad -2 \quad 1$$

Division

$$8 \overline{)\,14}\ 1 \quad -8 \quad 6$$

$14 / 8 = 1$

$= 1.75$

Division

modulo (%)

$3 \% 2 = 1$

$-3 \% 2 = -1$

$$2\overline{)3}\ 1,\quad -2,\quad 1 \text{ — modulo}$$

```
int main () {
    int a = 30;
    int b = 45;
    printf ("%d", b%a);
        return 0;
}
```

output : 15

$$30\overline{)45}\ 1,\quad -30,\quad 15 \text{ — modulo}$$

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Even number

$$4 \overline{)16} \quad 4$$
$$16$$
$$0$$

└ modulo

$x = 4 + 9 * 10$

$(4+9) * 10$

$13 * 10$

130 ✗

$4 + (9 * 10)$

$= 94$ ✓

$x = [4 * 3] / 6 * 2$

$= 12 \boxed{12/6} * 2$

$= 2$

$[2 * 2]$

$x = 4$

Precedence low

Left to Right

Relational Operators in C

Compare equal

Operator Symbol	Operators name
==	Equal to
!=	not equal to
<	less than
>	Greater than
<=	less then equal
>=	greater the equal
!=	not equal to

Exa:

Dad's age > my age

T(1)

A != a

8 ≥ 10

F(0)

Output always

T or F

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Logical Operators References

&& AND

true && true	=	true
T && F	=	false
F && T	=	False
F && F	=	False

OR ||

T || T	=	True
T || F	=	True
F || T	=	True
F || F	=	False

! NOT

! True = False

! False = True

Exor ^

T T = F

T F = T

F T = T

F F = F

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Assignment operator

- An assignment operator is used for assigning a value to a variable. The most common assignment operator.

Operator	Example	Same as
=	a=b	a=b
+=	a+=b	a= a+b
-=	a-=b	a= a-b
*=	a*=b	a= a*b
/=	a/=b	a= a/b
%=	a%=b	a= a%b

and

what was different

relation operator

Assignment operator

Example

$X_{30} == Y_{30}$ (T)

$X_{30} == Y_{21}$ (F)

Example

direct

Age = 19

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+=

Age = 19

Then, Age = 19+1

Age += 1

Sum = 2

আগের থেকে set করা হয়!

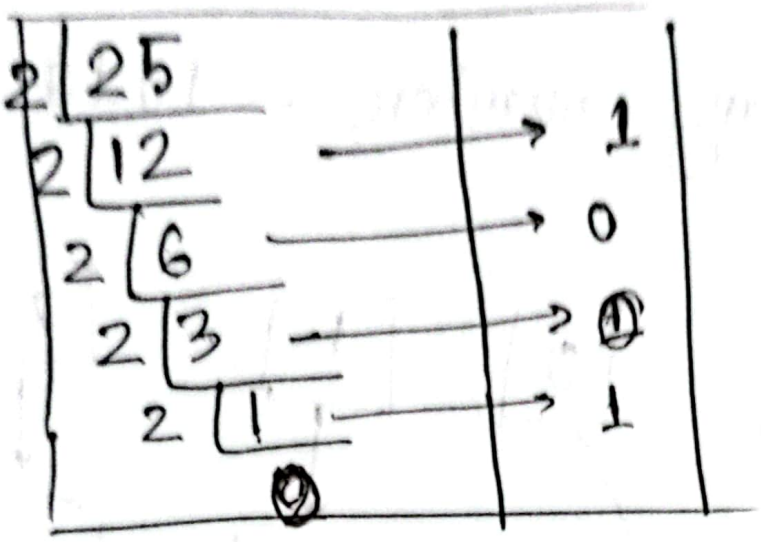
Sum = 2+4

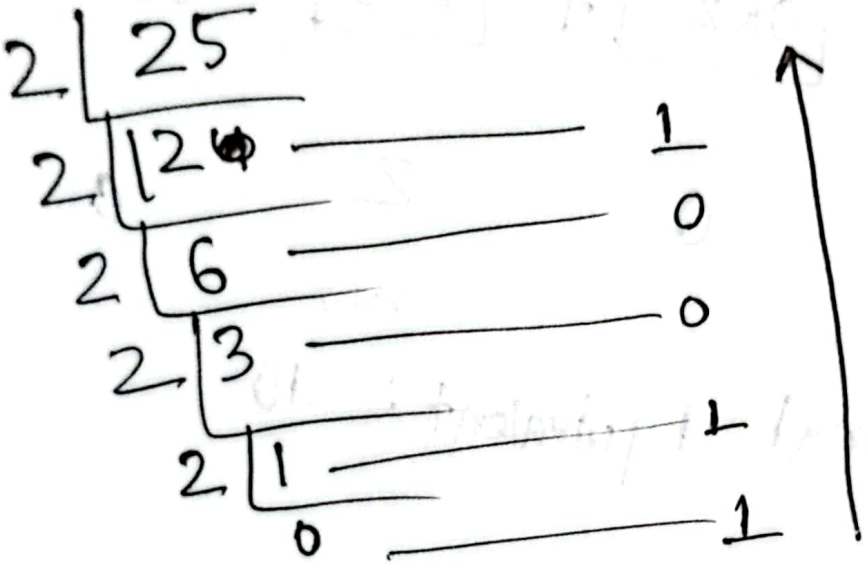
Sum += 4

Sum = 0

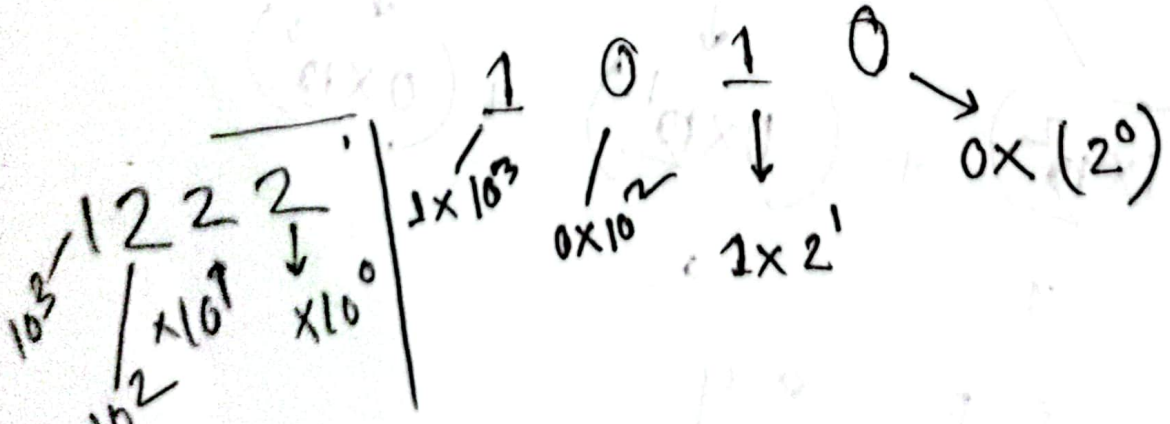
Sum += 1

Sum = 1





Binary number = 11001



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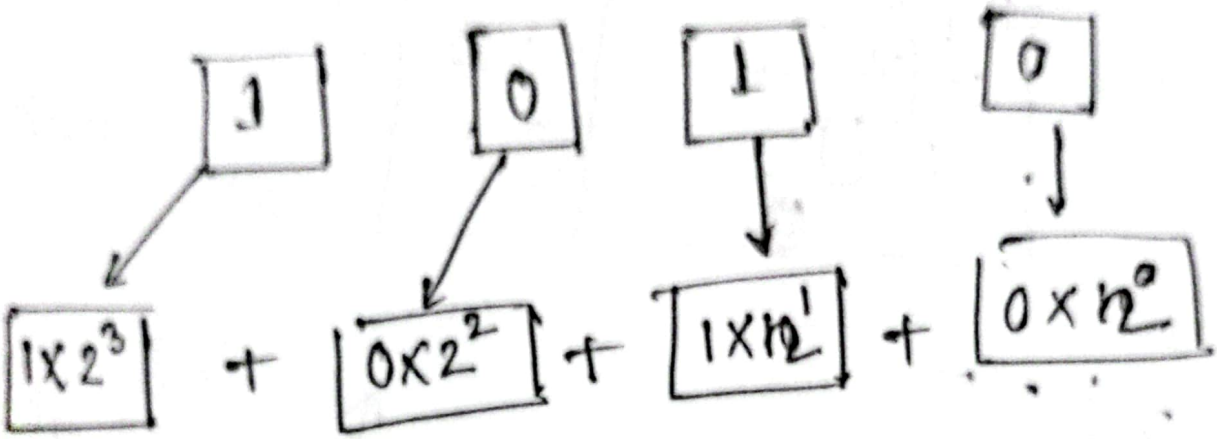
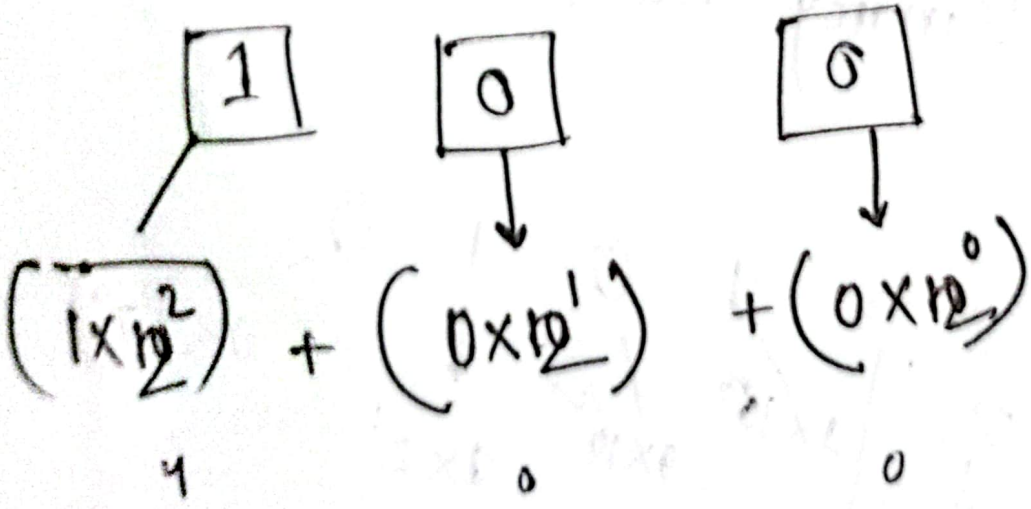
Binary number - 1010



Decimal Equivalent : 10

100



DE = 04

Bitwise operators = single &, | use mgt

&	|	^
and	or	xor

Double And

&&

&=?

0	1	0	1
0	0	0	1
F(0)	F(0)	F(0)	T(1)

0101 & 0001 = 0001

1101

|=? 1111

1111

1101 | 1111 = 1111

1101

^=? 0001

TTFF

1100

(1101 ^ 0001 = 1100

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IF Else

```
#include <stdio.h>
int main () {

if () {


}

else {

}

return 0;

}
```

// Robin sir class rules //

/* 10 = Argentina fan

else = Brazil fan */

```
# include <stdio.h>
  int main () {
     if (n == 10)
  int n;
  scanf ("%d", &n);
  if (n == 10) {
        printf ("Argentina Fan");
  }
  else (n!)
  else (n != 10) {
  printf ("Bra zil fan");
  }
  return 0;
  }
```

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